

NEURAL BELLMAN OPERATORS: RESPONSE TO THE REFEREE

1. REVISION AND SOURCE RECORD

We thank the referee for distinguishing the reliability of the numerical record from the economic and methodological contribution. This revision responds to the advisory report dated 3 October 2026 on the R8 candidate. The report is pinned at commit 3fb6d5bd; the reviewed candidate is f9a7e32e. We build on the subsequent complete R9 evidence commit f8215d6d, rather than discarding its repairs or reverting to an earlier manuscript. The advisory report was commissioned by the repository owner and is not an editorial decision of the Econometric Society.

The title, economic topic, and original applications remain unchanged. The complete R9 article and supplement are archived exactly; all inherited code, data, unsuccessful runs, and earlier review materials remain unchanged on the new branch. The root article is integrated rather than replaced by a detached research protocol. The new material is a continuous-time policy-performance theorem for the original dense nonlinear capital economy, its executable arithmetic verification, and a complete neural and direct-method study through dimension fifty.

The theorem compares each certified policy with all original admissible adapted controls in the primitive box, not only with policies in a neural class. A schedule-centred output map limits the neural approximation class and allows the performance bound to be computed from utility curvature, the verified coupling norm, and scalar quadrature. The comparison economy is not restricted to that output tube. The feasible schedule remains an explicit, cheaper benchmark with a tighter bound. This distinction prevents an architectural guarantee from being presented as evidence that neural training itself created the guarantee.

2. BLOCKING COMMENTS

2.0.0.1. *B1: Identify the actor's contribution rather than attribute search to it.* The new actor is an actual two-hidden-layer network deployed directly, with a scalar inward schedule guard but no local maximizing search, stored nodal policy, or maximizing labels. Its critic and actor are trained by separate derivative graphs; the independent-probe objective is used in the fit. Direct-tube and direct-full HJB

comparators exploit the same known global Hamiltonian solver. The untrained common schedule is also included. All 27 fits and their costs are reported. The theorem guarantees the schedule-centred policy class already at initialization; only the separately reported paired-payoff comparison concerns a gain from training. We do not attribute the historical hybrid results to raw actors or claim an advantage that the anchor comparison has not established.

2.0.0.2. B2: Make the certificate operational at a stated accuracy and total cost. R9's signed cover already recertifies the unchanged frozen preference policies and retains unresolved upper bounds. The present addition follows a different computational route within the original capital application: a positive interval factorization verifies a coupling-norm majorant, and outward scalar quadrature evaluates the continuous-time regret bound. No high-dimensional state covering is involved. Matrix verification and integral replay time are charged in full to each certified comparison; deployment time includes the scalar guard. The paper states the operation count of this verification calculation, without assigning that count to neural training or claiming a worst-case bound on spectral-majorant retries.

2.0.0.3. B3: Distinguish nodal refinement from a continuous-economy guarantee. The complete R9 full-domain frozen-policy refinement record remains in the paper, including the loss exceeding 2.12 away from the centre. Its guarded nested-grid training and separate fine-grid certificates are preserved. The new theorem does not reinterpret those nodal arrays as a diffusion bound. Instead, it provides a directly continuous-time result for the original capital SDE on the entire state space, uniformly over initial means and a stated range of initial dispersion. The preference model's continuously monitored stopping and state/time transfer defects remain separate from this result. This addresses the missing continuous-economy theorem without concealing the distinct unresolved transfer problem.

2.0.0.4. B4: Retain strong alternatives and compare the same economic problem. All classical, projection, actor-free, and direct comparisons from R8 and R9 remain. In the new study the direct-tube comparator uses the identical critic and admissible approximation tube, while direct-full searches the original complete action

interval. Both use the global scalar consistency equation rather than an intentionally weak optimizer. Samples, critic updates, seeds, and trace banks are matched; elapsed time is measured rather than declared equal. The analytical schedule receives its own row and is not charged a fictitious training cost. The table therefore exposes both any deployment saving of an actor and the stronger structural guarantee of the schedule.

2.0.0.5. *B5: Give a substantive theorem connected to the executed method.* The new theorem bounds the gain from any adapted deviation in the full capital economy. The proof first establishes an exact discounted mean-state identity, then combines a sharp log-utility tangent remainder with synchronous state stability. Completion of a quadratic deviation bound yields an explicit integral for the schedule's regret. A separate, two-sided tube-loss argument transfers the guarantee to the deployed neural policy. Every quantity is calculated with recorded outward arithmetic. This is not a convergence assertion for Adam, a restatement of an unknown transfer defect, or an assumed uniform neural residual. The mathematical ingredients are identified explicitly; the contribution is their computable model-specific performance consequence, not a claim that comparison or Gronwall inequalities are new.

2.0.0.6. *B6: Connect high dimension, neural training, and independent economic accuracy.* The new study has ten-, twenty-, and fifty-dimensional dense nonlinear economies, actual multilayer actor-critic training, strong direct comparators, all seed-level records, and a continuous-time policy bound in the same experiment. The bound covers arbitrary initial means and initial cross-sectional standard deviations up to the specified limit. It is independent of state-grid size and does not count an Euler or sampled-HJB diagnostic as proof. The original ten- and twenty-dimensional primitives and matrices are checked against the inherited implementation. The fifty-dimensional economy uses the same coupling-generation rule. No Riccati solve, separable reduction, or exact optimal parametric family is used.

2.0.0.7. *B7: Organize the argument around a complete chain without discarding the applications.* The new sections present the economic model, theorem, construc-

tive actor restriction, arithmetic implementation, training comparison, and welfare interpretation consecutively. Earlier recursive-utility, endogenous-preference, temporal-self, dynamic-game, viscosity, and trace results retain their places and labels. The added continuous-capital chain supplies depth within the existing NBO subject rather than replacing it with a different paper. The abstract and conclusion distinguish this new guarantee from the preceding nodal and simulation evidence.

3. MAJOR COMMENTS

3.0.0.1. *M1: Title and abstract.* The title remains *Neural Bellman Operators*. The abstract states the verified capital-policy result, the restricted approximation architecture, and the unrestricted economic comparison class. It also states that the schedule is cheaper and has the tighter structural bound. It makes no general actor-superiority claim.

3.0.0.2. *M2: Full-domain statistics.* The R9 maxima, quantiles, worst locations, boundary summaries, and saturation rates remain available in the integrated supplement and machine-readable record. The present capital theorem is uniform over the stated initial-state class, not just the mean of a validation sample. Held-out neural residuals are still labelled as diagnostics.

3.0.0.3. *M3: Economic interpretation of accuracy.* For the logarithmic-flow capital model, an externally financed proportional flow-consumption supplement has exactly known utility increment. We report an outward sufficient compensation percentage obtained from the regret bound. Financing, adjustment costs, state law, and terminal payoff are held as explicitly stated. This conversion is not exported to the endogenous-preference economy, which retains its own R9 compensation calculation.

3.0.0.4. *M4: Total cost at achieved accuracy.* The cost table reports actual training time, training plus the complete structural verifier, and full deployed-policy evaluation time. The verifier is charged without amortization in this table. All bi-section and state-query counts are recorded. Since the structural certificate exists before training, the paper does not label training time as time to first certification.

Common Brownian simulations are diagnostic research costs and are separately recorded, not required by the proof.

3.0.0.5. *M5: Action domains are economic primitives.* The capital economy retains its original $[.02, 2]^d$ primitive action box. The actor tube is an approximation restriction, not a new primitive or a restriction on competitors in the theorem. The preference action box and its binding-bound sensitivity remain conditional economic specifications as before.

3.0.0.6. *M6: Deployment object.* The deployed object is a saved network, the analytic common schedule, and the scalar inward arithmetic guard. It uses neither an off-grid interpolation of a policy table nor an online Bellman search. The direct comparators have distinct deployment routines and costs. Raw proposal timing is retained, but the main table reports the complete guarded deployment.

3.0.0.7. *M7: Dynamic games.* The finite-game best-response calculations and their state/action/time scope are preserved. The capital theorem is not used to upgrade them to continuous-game equilibria. The existing strategic section continues to distinguish unilateral dynamic deviations from a jointly differentiated actor objective.

3.0.0.8. *M8: A decision-relevant high-dimensional benchmark.* The schedule has its own proved regret bound against the continuous-time optimum. The actor's bound is obtained by adding a rigorously controlled tube allowance, rather than by comparing two unknown policies alone. Common-path comparisons separately report what the trained actor gains or loses relative to that schedule and to both direct alternatives. Unknown Euler bias is not subtracted from the analytical bound.

3.0.0.9. *M9: Conditional interval terminology.* All results retain the inherited outward binary64 execution contract. Polynomial transcendental bounds, checked square roots, interval spectral pivots, and raw replay checks are documented. No claim of a machine-formal proof or verified interpreter is made. The scalar deployment guard avoids assuming that a neural-library transcendental call itself supplies an outward enclosure.

3.0.0.10. *M10: Scientific architecture and preserved history.* The root paper integrates the new theorem and experiment before the conclusion, with the full proof in the appendix. Exact earlier roots and every prior evidence file are preserved. The current source, execution, table, and evidence manifests identify the new contribution independently of previous workflow success. The reply does not treat a successful build as referee acceptance or as proof of every desired numerical advantage.

4. REPRODUCTION AND REMAINING DISTINCTIONS

The source branch pins the entire program before remote execution. Its workflow reruns the new algebra, interval, gradient-routing, model-identity, and deployment tests; replays the inherited test suites without changing historical evidence; computes every declared certificate and fit; audits raw pathwise arrays and weight hashes; generates the tables; and compiles the article, supplement, and this response. Only the complete execution is committed as the evidence and referee copy. Failed commands are retained in logs and do not become successful evidence by omission.

The distinctions remaining after this addition are scientific rather than hidden accounting conventions: a structural policy guarantee is not a training convergence rate; a continuous-capital theorem is not a continuous-stopping theorem for endogenous preferences; and a trained actor must earn any claimed advantage over the feasible schedule in the measured comparison. The revised paper states its positive results at those precise levels while preserving the original research programme.